Multiple IV Infusions & Multiple Oral Practice 1

Module 6 - Multiple Dosings · Daily practice sheet · No date

Name _____ Date _____

1a 1. An 800-mg dose of a medication was administered as an IV infusion over a period of 3 hours to an 80-kg patient. Six hours after the start of the first infusion, a second of 800-mg dose was infused, again over 3 hours. The drug has an apparent volume of distribution that is 25% of body weight and a total body clearance is 4.62 L/hr. a. What is the plasma drug concentration at the end of the first infusion?

1b 1. An 800-mg dose of a medication was administered as an IV infusion over a period of 3 hours to an 80-kg patient. Six hours after the start of the first infusion, a second of 800-mg dose was infused, again over 3 hours. The drug has an apparent volume of distribution that is 25% of body weight and a total body clearance is 4.62 L/hr. b. What is the plasma drug concentration 6 hours after the cessation of the second infusion?

2a 2. An adult male (80 kg) was given 500 mg of an antibiotic orally every 12 hours for 2 weeks. The literature reports that the antibiotic is about 90% bioavailable and has a V_D of 0.24 L/kg. The elimination half-life is about 7.3 hours and the absorption half-life is about 75 minutes. Calculate a. Time required to reach the maximum concentration of the first dose.

2b 2. An adult male (80 kg) was given 500 mg of an antibiotic orally every 12 hours for 2 weeks. The literature reports that the antibiotic is about 90% bioavailable and has a V_D of 0.24 L/kg. The elimination half-life is about 7.3 hours and the absorption half-life is about 75 minutes. Calculate b. Maximum plasma drug concentration of the first dose.

2c 2. An adult male (80 kg) was given 500 mg of an antibiotic orally every 12 hours for 2 weeks. The literature reports that the antibiotic is about 90% bioavailable and has a V_D of 0.24 L/kg. The elimination half-life is about 7.3 hours and the absorption half-life is about 75 minutes. Calculate c. Minimum plasma drug concentration of the first dose.

2d 2. An adult male (80 kg) was given 500 mg of an antibiotic orally every 12 hours for 2 weeks. The literature reports that the antibiotic is about 90% bioavailable and has a V_D of 0.24 L/kg. The elimination half-life is about 7.3 hours and the absorption half-life is about 75 minutes. Calculate d. Area under the curve of the first dose.

2e 2. An adult male (80 kg) was given 500 mg of an antibiotic orally every 12 hours for 2 weeks. The literature reports that the antibiotic is about 90% bioavailable and has a V_D of 0.24 L/kg. The elimination half-life is about 7.3 hours and the absorption half-life is about 75 minutes. Calculate e. Time required to reach the maximum concentration at steady state.

2f 2. An adult male (80 kg) was given 500 mg of an antibiotic orally every 12 hours for 2 weeks. The literature reports that the antibiotic is about 90% bioavailable and has a V_D of 0.24 L/kg. The elimination half-life is about 7.3 hours and the absorption half-life is about 75 minutes. Calculate f. Maximum plasma drug concentration at steady state.

2g 2. An adult male (80 kg) was given 500 mg of an antibiotic orally every 12 hours for 2 weeks. The literature reports that the antibiotic is about 90% bioavailable and has a V_D of 0.24 L/kg. The elimination half-life is about 7.3 hours and the absorption half-life is about 75 minutes. Calculate g. Minimum plasma drug concentration at steady state.

2h 2. An adult male (80 kg) was given 500 mg of an antibiotic orally every 12 hours for 2 weeks. The literature reports that the antibiotic is about 90% bioavailable and has a V_D of 0.24 L/kg. The elimination half-life is about 7.3 hours and the absorption half-life is about 75 minutes. Calculate h. Average plasma drug concentration at steady state.

2i 2. An adult male (80 kg) was given 500 mg of an antibiotic orally every 12 hours for 2 weeks. The literature reports that the antibiotic is about 90% bioavailable and has a V_D of 0.24 L/kg. The elimination half-life is about 7.3 hours and the absorption half-life is about 75 minutes. Calculate i. Dose required to reach an average steady state conc of 25 mg/L.

2j 2. An adult male (80 kg) was given 500 mg of an antibiotic orally every 12 hours for 2 weeks. The literature reports that the antibiotic is about 90% bioavailable and has a V_D of 0.24 L/kg. The elimination half-life is about 7.3 hours and the absorption half-life is about 75 minutes. Calculate j. Average plasma drug concentration at steady state if the dose were increased to 750 mg.

2k 2. An adult male (80 kg) was given 500 mg of an antibiotic orally every 12 hours for 2 weeks. The literature reports that the antibiotic is about 90% bioavailable and has a V_D of 0.24 L/kg. The elimination half-life is about 7.3 hours and the absorption half-life is about 75 minutes. Calculate k. Average plasma drug concentration at steady state if the dosing interval was decreased to 8 hours (dose 500 mg).

Solutions

Her key: the answer and working on her own solved sheet or annotated slide. Worked by the drill: the drill's own working, where her sheet gives no worked value.

1a Her key

Answer: 28.86 mg/L

WHAT THE STEM GIVES

D_0	800-mg	$R = \frac{800}{3} = 266.67$ mg/hr
infusion time	3 hours	the denominator of R and the t in $(1 - e^{-kt})$
weight	80-kg	$V_D = 0.25 \times 80 = 20$ L
second start	Six hours after the start	*not needed: only the first infusion is asked*
V_D	25% of body weight	20 L; $k = \frac{4.62}{20} = 0.231$ hr^{-1}
Cl_T	4.62 L/hr	the denominator of the plateau $\frac{R}{Cl_T}$

THE WORKING, ONE LINE AT A TIME

UNIT

$$R = \frac{800 \text{ mg}}{3 \text{ hr}} = 266.67 \text{ mg/hr}; \; V_D = (0.25 \text{ L/kg})(80 \text{ kg}) = 20 \text{ L}; \; k = \frac{4.62 \text{ L/hr}}{20 \text{ L}} = 0.231 \text{ hr}^{-1}$$

The infusion rate is the dose over the infusion time; the volume is 25% of body weight; and k is clearance over volume, L/hr over L leaving hr^{-1}.

SETUP

$$C_p = \frac{R}{Cl_T}(1 - e^{-kt}), \text{ with } t = 3 \text{ hr}$$

During an infusion the concentration rises towards R over Cl_T; after t hours it has reached the fraction $(1 - e^{-kt})$ of that plateau. t is the infusion time, not the interval.

ALGEBRA

$$C_p = \frac{266.67 \text{ mg/hr}}{4.62 \text{ L/hr}}(1 - e^{-(0.231)(3)}) = (57.72 \text{ mg/L})(0.5) = 28.86 \text{ mg/L}$$

mg/hr over L/hr leaves mg/L; 3 hours is one half-life ($t_{½}$ = 0.693 over 0.231 = 3 hr), so the infusion stops at half of the 57.72 mg/L plateau.

> **DOES THE NUMBER MAKE SENSE?**
> The plateau R over Cl_T is 266.67 over 4.62, about 57.72 mg/L; the half-life is 0.693 over 0.231 = 3 hr, so a 3 hour infusion reaches exactly half the plateau: 28.86 mg/L. The answer must lie below 57.72 mg/L.

1b Her key

Answer: 9.02 mg/L

WHAT THE STEM GIVES

D_0	800-mg	each infusion ends at 28.86 mg/L, as in part a
infusion time	3 hours	sets the end times, 3 and 9 hr
weight	80-kg	$V_D = 0.25 \times 80 = 20$ L
second start	Six hours after the start	the second infusion runs from 6 to 9 hr
V_D	25% of body weight	20 L; $k = \frac{4.62}{20} = 0.231\ hr^{-1}$
Cl_T	4.62 L/hr	with V_D gives $k = 0.231\ hr^{-1}$
t	6 hours after the cessation	the second has declined 6 hr, the first 12 hr

THE WORKING, ONE LINE AT A TIME

SETUP

Time line: 0, 3, 6, 9, 15 hr

The first infusion runs from 0 to 3 hours; the second starts at 6 and ends at 9 hours; 6 hours after its end is 15 hours. The first infusion has then been declining for 12 hours and the second for 6.

SETUP

$$C_p = (28.86\ mg/L)\ e^{-k(6)} + (28.86\ mg/L)\ e^{-k(12)}$$

Each infusion contributes its end-of-infusion concentration, 28.86 mg/L, declined for the time since its own end; the two add because elimination is first order.

ALGEBRA

$$C_p = (28.86)(0.25) + (28.86)(0.0625) = 7.215 + 1.804 = 9.02\ mg/L$$

6 hours is two half-lives, leaving one quarter of the second infusion; 12 hours is four half-lives, leaving one sixteenth of the first; the exponents are pure numbers, so the unit stays mg/L.

> **DOES THE NUMBER MAKE SENSE?**
> Six hours is two half-lives, so a quarter of the second infusion remains, 28.86 × 0.25 = 7.215 mg/L; twelve hours is four half-lives, leaving a sixteenth of the first, 1.804; the sum, 9.02 mg/L, lies below 28.86.

2a Her key

Answer: 3.84 hr

WHAT THE STEM GIVES

weight	80 kg	*not needed*
D_0	500 mg	*not needed: t_{max} depends on k_a and k alone*
τ	every 12 hours	*not needed for the first-dose peak*
duration	2 weeks	*not needed: long enough for steady state*

F	90% bioavailable	*not needed*
V_D	0.24 L/kg	*not needed*
t½	7.3 hours	gives $k = \frac{0.693}{7.3} = 0.0949\ hr^{-1}$
t½a	75 minutes	1.25 hr, so $k_a = \frac{0.693}{1.25} = 0.5544\ hr^{-1}$

THE WORKING, ONE LINE AT A TIME

UNIT

$$k_a = \frac{0.693}{1.25\ hr} = 0.5544\ hr^{-1};\ k = \frac{0.693}{7.3\ hr} = 0.0949\ hr^{-1}$$

The absorption half-life, 75 minutes, is 1.25 hours; each rate constant is 0.693 over its own half-life, in reciprocal hours.

SETUP

$$t_{max} = \frac{\ln(k_a/k)}{k_a - k}$$

The time to peak after one oral dose depends only on the two rate constants; the dose, F and the volume do not enter.

ALGEBRA

$$t_{max} = \frac{\ln(0.5544/0.0949)}{(0.5544 - 0.0949)\ hr^{-1}} = \frac{1.765}{0.4595\ hr^{-1}} = 3.84\ hr$$

The ratio inside the logarithm, 5.84, is a pure number, and dividing its logarithm, 1.765, by (k_a − k) in hr^{-1} leaves hours: the first-dose peak comes 3.84 hours after the dose.

> **DOES THE NUMBER MAKE SENSE?**
> Absorption (half-life 1.25 hr) is about six times faster than elimination (7.3 hours), so the peak comes after a few absorption half-lives but well within one elimination half-life; 3.84 hr fits.

2b Her key

Answer: 16.28 mg/L

WHAT THE STEM GIVES

weight	80 kg	$V_D = 0.24 \times 80 = 19.2$ L
D_0	500 mg	the dose in the numerator
τ	every 12 hours	*not needed for the first-dose peak*
duration	2 weeks	*not needed: long enough for steady state*
F	90% bioavailable	F = 0.9 in the numerator
V_D	0.24 L/kg	times 80 kg gives 19.2 L, the denominator
t½	7.3 hours	gives $k = \frac{0.693}{7.3} = 0.0949\ hr^{-1}$
t½a	75 minutes	1.25 hr, so $k_a = \frac{0.693}{1.25} = 0.5544\ hr^{-1}$

THE WORKING, ONE LINE AT A TIME

SETUP

$$C_{max} = \frac{F\,k_a\,D_0}{V_D(k_a - k)}\,(e^{-k\,t_{max}} - e^{-k_a\,t_{max}}),\ V_D = (0.24\ \text{L/kg})(80\ \text{kg}) = 19.2\ \text{L}$$

C_{max} is the single-dose oral equation evaluated at t_{max} = 3.84 hr, with k = 0.0949 hr^{-1} and k_a = 0.5544 hr^{-1}; the volume per kilogram is multiplied by the 80 kg weight first.

ALGEBRA

$$C_{max} = \frac{(0.9)(500\ \text{mg})(0.5544\ \text{hr}^{-1})}{(19.2\ \text{L})(0.5544 - 0.0949)\ \text{hr}^{-1}}\,(e^{-(0.0949)(3.84)} - e^{-(0.5544)(3.84)})$$

Reciprocal hours cancel between k_a in the numerator and (k_a − k) in the denominator, and mg over L leaves mg/L; the coefficient in front of the brackets is 28.28 mg/L.

ALGEBRA

$$C_{max} = (28.28\ \text{mg/L})(0.6945 - 0.1190) = 16.28\ \text{mg/L}$$

At the peak the elimination term is 0.6945 and the absorption term 0.1190; their difference times the coefficient is the first-dose peak.

> **DOES THE NUMBER MAKE SENSE?**
> The absorbed dose, 0.9 × 500 mg, in 19.2 L would give about 23 mg/L if it were all in the body at once; the peak must lie below that, and 16.28 mg/L does, about two thirds of it because elimination runs during absorption.

2c Her key

Answer: 9.02 mg/L

WHAT THE STEM GIVES

weight	80 kg	$V_D = 0.24 \times 80 = 19.2$ L
D_0	500 mg	the dose in the numerator
τ	every 12 hours	t = τ = 12 hr in both exponents
duration	2 weeks	*not needed: long enough for steady state*
F	90% bioavailable	F = 0.9 in the numerator
V_D	0.24 L/kg	times 80 kg gives 19.2 L, the denominator
t½	7.3 hours	gives $k = \frac{0.693}{7.3} = 0.0949\ hr^{-1}$
t½a	75 minutes	1.25 hr, so $k_a = \frac{0.693}{1.25} = 0.5544\ hr^{-1}$

THE WORKING, ONE LINE AT A TIME

SETUP

$$C_{min} = \frac{F\,k_a\,D_0}{V_D(k_a - k)}\,(e^{-k\tau} - e^{-k_a\,\tau}),\ \text{with}\ \tau = 12\ \text{hr}$$

The trough of the first dose is the same oral equation at the end of the dosing interval, just before the second dose.

ALGEBRA

$C_{min} = (28.28\ mg/L)(e^{-(0.0949)(12)} - e^{-(0.5544)(12)}) = (28.28\ mg/L)(0.3202 - 0.0013) = 9.02\ mg/L$

By 12 hours absorption is essentially complete ($e^{-6.65} = 0.0013$), so the trough is set by the elimination term, 0.3202, times the coefficient.

> **DOES THE NUMBER MAKE SENSE?**
> Twelve hours is more than one elimination half-life of 7.3 hours and absorption is over, so the trough is below half the first-dose peak; the coefficient 28.28 mg/L times 0.3202 is about 9 mg/L, so 9.02 fits.

2d

Her key

Answer: 246.97 (mg/L)hr

WHAT THE STEM GIVES

weight	80 kg	$V_D = 0.24 \times 80 = 19.2$ L
D_0	500 mg	the dose in the numerator
τ	every 12 hours	*not needed: the area of one dose has no interval*
duration	2 weeks	*not needed: long enough for steady state*
F	90% bioavailable	F = 0.9 in the numerator
V_D	0.24 L/kg	times 80 kg gives 19.2 L, the denominator
t½	7.3 hours	gives $k = \frac{0.693}{7.3} = 0.0949\ hr^{-1}$
t½a	75 minutes	*not needed: the area does not depend on k_a*

THE WORKING, ONE LINE AT A TIME

SETUP

$$AUC = \frac{FD_0}{V_D\,k}$$

The area under one dose is the absorbed dose over the clearance, and clearance is $V_D \times k$; only the absorbed fraction, 0.9 of 500 mg, reaches the plasma.

ALGEBRA

$$AUC = \frac{(0.9)(500\ mg)}{(19.2\ L)(0.0949\ hr^{-1})} = \frac{450\ mg}{1.822\ L/hr} = 246.97\ (mg/L)hr$$

mg over L/hr gives (mg/L)hr, the unit of an area under a concentration–time curve; $V_D \times k = 1.822$ L/hr is the clearance, and 450 mg absorbed over it is the area of one dose.

> **DOES THE NUMBER MAKE SENSE?**
> Clearance $V_D \times k$ is 1.822 L/hr and 450 mg is absorbed, so the area is 450 over 1.822, near 250 (mg/L)hr; 246.97 fits, and using 500 mg instead of 450 would give a value a ninth too high.

Her key carries k as 0.0949 hr^{-1}; with the unrounded 0.693 over 7.3 the area is 246.89 (mg/L)hr. Either is accepted.

2e Her key

Answer: 3 hr

WHAT THE STEM GIVES

weight	80 kg	*not needed*
D_0	500 mg	*not needed: $t_{max}\infty$ depends on k_a, k and τ alone*
τ	every 12 hours	in both $(1 - e^{-k\tau})$ and $(1 - e^{-k_a \tau})$
duration	2 weeks	*not needed: long enough for steady state*
F	90% bioavailable	*not needed*
V_D	0.24 L/kg	*not needed*
t½	7.3 hours	gives $k = \frac{0.693}{7.3} = 0.0949\ hr^{-1}$
t½a	75 minutes	1.25 hr, so $k_a = \frac{0.693}{1.25} = 0.5544\ hr^{-1}$

THE WORKING, ONE LINE AT A TIME

SETUP

$$t_{max}\infty = \frac{1}{k_a - k} \ln\left[\frac{k_a(1 - e^{-k\tau})}{k(1 - e^{-k_a \tau})}\right]$$

At steady state the interval enters the time to peak through the two bracketed terms, which is what makes it differ from the single-dose t_{max}.

ALGEBRA

$$1 - e^{-(0.0949)(12)} = 0.6798;\ 1 - e^{-(0.5544)(12)} = 0.9987$$

Over one 12-hour interval 68% of the drug in the body is eliminated, and absorption of a dose is complete (0.9987); these two fractions carry the interval into the time to peak.

ALGEBRA

$$t_{max}\infty = \frac{1}{0.4595\ hr^{-1}} \ln\left[\frac{(0.5544)(0.6798)}{(0.0949)(0.9987)}\right] = (2.176\ hr)(\ln 3.976) = 3\ hr$$

The ratio inside the logarithm, 3.976, is smaller than the single-dose ratio k_a over k, 5.84, so the steady-state peak comes earlier: 3 hours against 3.84.

> **DOES THE NUMBER MAKE SENSE?**
> The steady-state peak comes earlier than the first-dose peak because drug from earlier doses is being eliminated while the new dose absorbs; the ratio inside the logarithm, 3.976, is below the single-dose ratio of k_a to k, so $t_{max}\infty$ is shorter than the part a value: 3 hr.

2f Her key

Answer: 25.94 mg/L

WHAT THE STEM GIVES

weight	80 kg	$V_D = 0.24 \times 80 = 19.2$ L
D_0	500 mg	the dose in the numerator

τ	every 12 hours	the accumulation factor $\frac{1}{1 - e^{-k\tau}} = 1.471$
duration	2 weeks	*not needed: long enough for steady state*
F	90% bioavailable	F = 0.9 in the numerator
V_D	0.24 L/kg	times 80 kg gives 19.2 L, the denominator
t½	7.3 hours	gives k = $\frac{0.693}{7.3} = 0.0949$ hr^{-1}
t½a	75 minutes	gave $t_{max}^{\infty} = 3$ hr in part e, the time in $e^{-k\, t_{max}^{\infty}}$

THE WORKING, ONE LINE AT A TIME

SETUP

$$C_{max}^{\infty} = \frac{FD_0}{V_D} \times \frac{1}{1 - e^{-k\tau}} \times e^{-k\, t_{max}^{\infty}}, \text{ with } t_{max}^{\infty} = 3 \text{ hr}$$

The steady-state peak is the absorbed dose over the volume, raised by the accumulation factor and declined for the time to the steady-state peak.

ALGEBRA

$$C_{max}^{\infty} = \frac{(0.9)(500 \text{ mg})}{19.2 \text{ L}} \times \frac{1}{1 - e^{-(0.0949)(12)}} \times e^{-(0.0949)(3)} = (23.44 \text{ mg/L})$$

$$(1.471)(0.7523) = 25.94 \text{ mg/L}$$

mg over L is mg/L; the accumulation factor 1.471 and the decline 0.7523 are pure numbers. The peak at steady state is above the first-dose peak of 16.28 mg/L.

> **DOES THE NUMBER MAKE SENSE?**
> 0.9 × 500 mg in 19.2 L is 23.44 mg/L; the accumulation factor 1.471 raises it and the 3 hour decline 0.7523 lowers it, so the steady-state peak is a little above 23.44 mg/L and above the first-dose peak: 25.94 fits.

Her key uses k 0.0949 hr^{-1} and t_{max}^{∞} 3 hr; the unrounded values give 25.92 mg/L. Either is accepted.

2g

Her key

Answer: 13.32 mg/L

WHAT THE STEM GIVES

weight	80 kg	$V_D = 0.24 \times 80 = 19.2$ L
D_0	500 mg	the dose in the numerator
τ	every 12 hours	the accumulation factor 1.471 and the decline $e^{-k\tau}$ over 12 hr
duration	2 weeks	*not needed: long enough for steady state*
F	90% bioavailable	F = 0.9 in the numerator
V_D	0.24 L/kg	times 80 kg gives 19.2 L, the denominator
t½	7.3 hours	gives k = $\frac{0.693}{7.3} = 0.0949$ hr^{-1}

t½a	75 minutes	1.25 hr, so $k_a = \frac{0.693}{1.25} = 0.5544\ hr^{-1}$

THE WORKING, ONE LINE AT A TIME

SETUP

$$C_{min}^{\infty} = \frac{F\,k_a\,D_0}{V_D(k_a - k)} \times \frac{1}{1 - e^{-k\tau}} \times e^{-k\tau}$$

The steady-state trough is the first-dose coefficient raised by the accumulation factor and declined for one full interval.

ALGEBRA

$C_{min}^{\infty} = (28.28\ mg/L)(1.471)\ e^{-(0.0949)(12)} = (28.28\ mg/L)(1.471)(0.3202) = 13.32\ mg/L$

The coefficient 28.28 mg/L is the one from part b; times the accumulation factor and the 12-hour decline it gives the trough, above the first-dose trough of 9.02 mg/L.

> **DOES THE NUMBER MAKE SENSE?**
> The steady-state trough is the first-dose trough times the accumulation factor 1.471, so about half as much again as the part c value, near 13 mg/L; 13.32 fits and lies below the steady-state peak.

With the unrounded rate constants the trough is 13.31 mg/L; her key prints 13.32. Either is accepted.

2h Her key

Answer: 20.58 mg/L

WHAT THE STEM GIVES

weight	80 kg	$V_D = 0.24 \times 80 = 19.2$ L
D_0	500 mg	the dose in the numerator
τ	every 12 hours	multiplies V_D and k in the denominator
duration	2 weeks	*not needed: long enough for steady state*
F	90% bioavailable	F = 0.9 in the numerator
V_D	0.24 L/kg	times 80 kg gives 19.2 L, the denominator
t½	7.3 hours	gives $k = \frac{0.693}{7.3} = 0.0949\ hr^{-1}$
t½a	75 minutes	*not needed: the average does not depend on k_a*

THE WORKING, ONE LINE AT A TIME

SETUP

$$C_{avg}^{\infty} = \frac{FD_0}{V_D\,k\,\tau}$$

The average over one interval at steady state is the absorbed dose per interval over the volume of plasma cleared per interval; it needs no t_{max} and no accumulation factor.

ALGEBRA

$$C_{avg}^{\infty} = \frac{(0.9)(500\text{ mg})}{(19.2\text{ L})(0.0949\text{ hr}^{-1})(12\text{ hr})} = \frac{450\text{ mg}}{21.86\text{ L}} = 20.58\text{ mg/L}$$

L × hr⁻¹ × hr leaves litres, so mg over L is mg/L; the average lies between the trough, 13.32, and the peak, 25.94 mg/L.

DOES THE NUMBER MAKE SENSE?

450 mg is absorbed every 12 hours and 19.2 L × 0.0949 hr⁻¹ × 12 hr = 21.86 L of plasma is cleared in that time, so the average is a little over 20 mg/L; it lies between the steady-state trough and peak.

With the unrounded k the average is 20.57 mg/L; her key prints 20.58. Either is accepted.

2i

Her key

Answer: 600 mg

WHAT THE STEM GIVES

weight	80 kg	$V_D = 0.24 \times 80 = 19.2$ L
D_0	500 mg	*not needed: replaced by the dose asked for*
τ	every 12 hours	multiplies V_D and k
duration	2 weeks	*not needed: long enough for steady state*
F	90% bioavailable	divides: $D_0 = \frac{C_{avg}^{\infty}\, V_D\, k\, \tau}{F}$
V_D	0.24 L/kg	times 80 kg gives 19.2 L, the denominator
t½	7.3 hours	gives $k = \frac{0.693}{7.3} = 0.0949\text{ hr}^{-1}$
t½a	75 minutes	*not needed*
C_{avg}^{∞}	25 mg/L	the target, multiplied by V_D, k and τ

THE WORKING, ONE LINE AT A TIME

SETUP

$$C_{avg}^{\infty} = \frac{FD_0}{V_D\, k\, \tau}, \text{ so } D_0 = \frac{C_{avg}^{\infty}\, V_D\, k\, \tau}{F}$$

The average equation is rearranged for the dose; dividing by F = 0.9 makes the oral dose larger than the amount that has to be absorbed, because only 90% of a dose reaches the circulation.

ALGEBRA

$$D_0 = \frac{(25\text{ mg/L})(19.2\text{ L})(0.0949\text{ hr}^{-1})(12\text{ hr})}{0.9} = \frac{546.6\text{ mg}}{0.9} = 607.36\text{ mg}$$

mg/L × L × hr⁻¹ × hr leaves mg: 546.6 mg must be absorbed per interval, and 607.36 mg must be given for 546.6 mg to be absorbed.

ROUND

$D_0 = 600$ mg

Her key rounds 607.36 mg to 600 mg, a dose that can be given; 600 mg every 12 hours gives an average close to the 25 mg/L target.

> **DOES THE NUMBER MAKE SENSE?**
> 500 mg every 12 hours gave an average of about 20 mg/L, so 25 mg/L needs about a quarter more, near 600 mg; 607.36 mg rounds to 600 mg. The oral dose is above the 546.6 mg that must be absorbed because F = 0.9.

Her key gets 607.36 mg with k 0.0949 hr^{-1} and reports 600 mg; the unrounded k gives 607.56 mg. 600 and 607 are both accepted.

2j Her key

Answer: 30.87 mg/L

WHAT THE STEM GIVES

weight	80 kg	$V_D = 0.24 \times 80 = 19.2$ L
D_0	500 mg	*not needed: replaced by 750 mg*
τ	every 12 hours	multiplies V_D and k in the denominator
duration	2 weeks	*not needed: long enough for steady state*
F	90% bioavailable	F = 0.9 in the numerator
V_D	0.24 L/kg	times 80 kg gives 19.2 L, the denominator
t½	7.3 hours	gives $k = \frac{0.693}{7.3} = 0.0949\ hr^{-1}$
t½a	75 minutes	*not needed*
D_0 new	750 mg	the dose in the numerator

THE WORKING, ONE LINE AT A TIME

SETUP

$$C_{avg}^{\infty} = \frac{FD_0}{V_D\, k\, \tau}, \text{ with } D_0 = 750 \text{ mg}$$

Only the dose changes; F, the volume, k and the interval are the same as in part h, so the average scales with the dose.

ALGEBRA

$$C_{avg}^{\infty} = \frac{(0.9)(750\ mg)}{(19.2\ L)(0.0949\ hr^{-1})(12\ hr)} = \frac{675\ mg}{21.86\ L} = 30.87\ mg/L$$

L × hr^{-1} × hr leaves litres, so mg over L is mg/L; 750 is 1.5 times 500, and the average is 1.5 times the 20.58 mg/L of part h.

> **DOES THE NUMBER MAKE SENSE?**
> Each 500 mg gave an average near 20 mg/L (part h), and 750 mg is half as much again, so near 31 mg/L; 30.87 fits. The average scales with the dose because clearance does not change.

With the unrounded k the average is 30.86 mg/L; her key prints 30.87. Either is accepted.

2k Her key

Answer: 30.87 mg/L

WHAT THE STEM GIVES

weight	80 kg	$V_D = 0.24 \times 80 = 19.2$ L
D_0	500 mg	the dose in the numerator
τ	every 12 hours	*not needed: replaced by 8 hours*
duration	2 weeks	*not needed: long enough for steady state*
F	90% bioavailable	F = 0.9 in the numerator
V_D	0.24 L/kg	times 80 kg gives 19.2 L, the denominator
t½	7.3 hours	gives $k = \frac{0.693}{7.3} = 0.0949\ \text{hr}^{-1}$
t½a	75 minutes	*not needed*
τ new	8 hours	multiplies V_D and k in the denominator

THE WORKING, ONE LINE AT A TIME

SETUP

$$C_{avg}^{\infty} = \frac{FD_0}{V_D\, k\, \tau}, \text{ with } \tau = 8 \text{ hr}$$

Only the interval changes; the same 450 mg is absorbed every 8 hours instead of every 12, so less plasma is cleared per dose and the average rises.

ALGEBRA

$$C_{avg}^{\infty} = \frac{(0.9)(500\ \text{mg})}{(19.2\ \text{L})(0.0949\ \text{hr}^{-1})(8\ \text{hr})} = \frac{450\ \text{mg}}{14.58\ \text{L}} = 30.87\ \text{mg/L}$$

The average rises in the ratio 12 over 8, from 20.58 to 30.87 mg/L, the same value as giving 750 mg every 12 hours, because the daily absorbed dose is the same, 1350 mg, in both regimens.

> **DOES THE NUMBER MAKE SENSE?**
> Shortening the interval from 12 to 8 hours raises the average in the ratio 12 to 8, half as much again as the part h value near 20 mg/L: near 31 mg/L; 30.87 fits because the same 450 mg is absorbed in fewer hours.

With the unrounded k the average is 30.86 mg/L; her key prints 30.87. Either is accepted.